

PlantVillage Dataset Preprocessing

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September 13, 2026

0.1 Dataset Preparation and Image Pairing

The PlantVillage dataset [?] contains three parallel representations of plant-leaf images: color, grayscale, and segmented images. For the segmentation experiment, images from the color directory were considered as the model inputs, while their corresponding images from the segmented directory were retained as the segmentation supervision source.

An initial dataset audit was performed before model development. The color and grayscale directories each contained 54,305 images, whereas the segmented directory contained 54,306 images across 38 classes. Class-wise comparison revealed one additional segmented sample in the `Grape__Esca_(Black_Measles)` class. The unmatched segmented sample was excluded from the valid-pair set, resulting in 54,305 corresponding color-segmented image pairs.

Because the naming convention was not completely uniform between the color and segmented directories, a two-stage image-pairing procedure was employed. First, normalized full filenames were used to establish direct correspondence after removing segmentation-specific filename suffixes. When a direct filename match was unavailable, a descriptor-based fallback procedure was applied after normalization of the identifier components. Using this procedure, 53,109 image pairs were matched through normalized full filenames and 1,196 pairs through the descriptor-based fallback strategy. The final pairing process produced 54,305 unique pairs with no unresolved color images and no duplicated color or segmented paths.

To make the experimental pipeline reproducible across different runtime sessions, only relative image paths were stored in the pairing manifest rather than environment-specific absolute paths.

0.2 Training, Validation, and Test Partitioning

The 54,305 paired samples were divided into training, validation, and test sets using a stratified 80:10:10 split with a fixed random seed of 42. Stratification was performed according to the PlantVillage class labels so that the class distribution was proportionally maintained across all three subsets. The resulting partition contained 43,444 training samples, 5,430 validation samples, and 5,431 testing samples.

The three partitions were checked for sample-path overlap, and no overlap was observed between the training and validation, training and test, or validation and test subsets. The resulting split was stored as a fixed manifest and is reused for all segmentation models to ensure that every architecture is trained and evaluated on exactly the same data partitions.

0.3 Segmentation Supervision Preparation

The segmented images supplied with the PlantVillage dataset were retained as the segmentation supervision source. No independent manually annotated mask dataset was introduced, and no separate collection of binary mask image files was stored. The color and segmented images are loaded as corresponding pairs during runtime.

Prior to model training, the paired images were visually inspected and a full-dataset quality-control procedure was incorporated into the pipeline. Spatial consistency, foreground extraction, and target validity are verified before the final segmentation target formulation is fixed for model training.

0.4 Preprocessing Framework

A common preprocessing framework was prepared for all segmentation models. Images are resized to 256×256 pixels. Training samples are subjected to synchronized spatial augmentation, including horizontal flipping, vertical flipping, and random 90° rotation, so that identical geometric transformations are applied to the input image and its corresponding segmentation target. Input RGB images are normalized using ImageNet normalization statistics. Validation and test samples are resized and normalized without stochastic augmentation.

The same dataset partition, preprocessing procedure, and evaluation protocol are maintained for all evaluated segmentation architectures to support a controlled and fair model comparison.

0.5 Dataset Auditing and Image Pairing

The PlantVillage dataset contains three parallel image representations: color, grayscale, and segmented images. An initial dataset audit was performed prior to the segmentation experiments to verify the number of samples and the correspondence between the color and segmented directories. A total of 38 classes were identified. The color and grayscale directories each contained 54,305 images, whereas the segmented directory initially contained 54,306 images.

A class-wise comparison identified one additional segmented image in the `Grape__Esca_(Black_Measles)` class. This unmatched segmented sample was removed from the valid-pair set, resulting in 54,305 color images and 54,305 corresponding segmented images.

Since the filename structure was not completely consistent between the color and segmented directories, a two-stage pairing procedure was applied. Normalized full filenames were first used to

establish direct correspondence. For samples whose filenames differed because of identifier formatting, a descriptor-based fallback matching procedure was subsequently applied. Using this approach, 53,109 image pairs were matched through normalized full filenames and 1,196 pairs through the descriptor-based fallback procedure. The final pairing stage produced 54,305 unique color-segmented pairs with no unresolved samples and no duplicate color or segmented paths.

Relative file paths were stored in a persistent pairing manifest rather than runtime-specific absolute paths to ensure that the same paired samples could be reconstructed across independent Google Colab sessions.

0.6 Training, Validation, and Test Partitioning

The 54,305 paired samples were partitioned using a stratified 80:10:10 training, validation, and test split with a fixed random seed of 42. Stratification was performed using the PlantVillage class labels to preserve the class distribution across the three subsets. The initial partition contained 43,444 training samples, 5,430 validation samples, and 5,431 test samples.

Sample-level overlap was explicitly examined between all subsets. No overlap was found between the training and validation, training and test, or validation and test sets. The resulting partition was stored as a fixed manifest so that all segmentation architectures can be trained and evaluated using exactly the same data partition.

0.7 Segmentation Target Preparation

The RGB images from the PlantVillage `color` directory were used as model inputs, while the corresponding dataset-provided segmented images served as the segmentation supervision source. To avoid introducing an independently annotated mask dataset, binary segmentation targets were generated dynamically during data loading rather than stored as separate mask files.

Following source-quality screening, the retained segmented images contained reliable near-black background regions. For each retained sample, pixels whose maximum RGB intensity exceeded a threshold of 15 were assigned to the foreground class, while near-black pixels were assigned to the background class. Thus, foreground and background were encoded as 1 and 0, respectively. For the small number of valid pairs having different spatial resolutions, the binary target was first derived at the native resolution of the segmented image and then spatially aligned with the corresponding input using nearest-neighbor interpolation to preserve discrete label values.

The complete target-generation procedure was validated across all 52,969 retained samples. No empty or effectively full invalid targets were detected after the final quality-control procedure. The resulting fixed segmentation manifest and preprocessing configuration were stored and reused across all segmentation architectures to ensure identical training, validation, and test data for fair model comparison.

0.8 Quality Control of Paired Segmentation Data

A quality-control procedure was performed on the paired color and dataset-provided segmented images prior to model training. Visual inspection confirmed the correspondence between randomly selected color and segmented samples. A full-dataset spatial-dimension check identified four pairs in which the color image had a resolution of 256×256 pixels while the corresponding segmented source had a different spatial resolution. Visual verification confirmed that these four cases represented the correct corresponding leaf samples; therefore, they were retained for subsequent spatial alignment rather than discarded.

Quality control also identified one unusable sample from the `Soybean_healthy` class, for which the original image did not contain a usable leaf region and the corresponding segmented image contained no foreground information. The correctly matched but unusable pair was therefore excluded. Additional source-quality screening was subsequently performed to ensure that only segmented images providing reliable foreground-background separation were used for segmentation supervision. Samples with ambiguous non-black backgrounds or insufficient explicit background separation were not included in the final segmentation experiment. This procedure resulted in a final segmentation dataset of 52,969 samples while retaining all 38 PlantVillage classes. The final fixed partition contained 42,371 training samples, 5,300 validation samples, and 5,298 test samples, with no overlap between the three subsets.